Supplement: Methods and Results for Direct Elicitation versus Ranked Nodes

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1 Construction and evaluation details

This section complements the main article's Method with the numerical specification and retrieval settings. It describes the original experimental pipeline, distinct from the alternative fits reported below.

1.1 Scrum reference reconstruction

The evaluated XML stores ranked-node functions, weights, variances, and expressions. We reconstructed its 249 conditional rows using equal-width intervals and five endpoint-inclusive points per parent state, preserving two piecewise expressions and the storage order of a volatility variable. The archived implementation has 42 nodes and three states, whereas the published framework describes 44 nodes and five states. The reconstructed tables are not an independently certified AgenaRisk export. Other references are stored CPTs; SESSI uses the original elicited table in Appendix A3 of its source publication.

1.2 Ranked-node equations and numerical fitting

Ordered states divide $[0, 1]$ into equal intervals. Negative influences reverse semantic rank before aggregation; predictions are mapped back to the stored child order. Anchors include all-low, all-high, all-middle (rank $\lfloor (c-1)/2 \rfloor$ for each parent's c states), each one-high/rest-low configuration, and each one-low/rest-high configuration, with duplicates removed. The anchor selection consults no reference probabilities.

For k parents, let G_r be the Cartesian product of five equally spaced, endpoint-inclusive points within each parent interval for row r. With standard-normal CDF Φ, child interval $[a_s, b_s]$, aggregation mean μ, and variance v, define

$$T_s(\mu, v) = \frac{\Phi((b_s - \mu)/\sqrt{v}) - \Phi((a_s - \mu)/\sqrt{v})}{\Phi((1-\mu)/\sqrt{v}) - \Phi(-\mu/\sqrt{v})}. \tag{1}$$

The predicted probability is $q_{rs} = 5^{-k} \sum_{x \in G_r} T_s(f(x; w), v)$. Each truncated distribution is normalized before averaging. The mean functions are

$$f_{\text{mean}}(x; w) = \frac{\sum_i w_i x_i}{\sum_i w_i}, \tag{2}$$

$$g_i(x; w) = \frac{w_i x_i + \sum_{j \neq i} x_j}{w_i + k - 1}, \tag{3}$$

$$f_{\min} = \min_i g_i, \qquad f_{\max} = \max_i g_i, \tag{4}$$

$$f_{\text{mix}}(x; a) = (1-a) \min_i x_i + a \max_i x_i. \tag{5}$$

For a single parent, the mean is its aligned value x_1. Weighted-mean log weight ratios relative to the last parent lie in $[-6, 6]$; weighted-minimum/maximum weights lie in $[1, 20]$; $a \in [0, 1]$; and $v \in [10^{-4}, 1]$. Variance and minimum/maximum weights use logarithmic parameterizations.

For the family selected by the LLM, the fitter minimized mean anchor total variation between fitted and elicited distributions. It used eight deterministic starts (seed 42), coordinate-search steps 2^{-2} through 2^{-11} on normalized parameter coordinates, at most 20 sweeps per step, and an improvement threshold of 10^{-12}. Ties retained the earlier

candidate. This seed controls fitting, not LLM sampling. The fitter received no reference CPT input. Poor but valid fits were retained; similar tables can arise from different parameters, so agreement does not establish parameter recovery.

1.3 Evidence retrieval and elicitation controls

Before collection, retrieval filtered evidence cards by model context and scored overlap between their variable tags and the target-plus-parent set; general-purpose tags contributed 0.25. It selected up to three positive-scoring cards, breaking ties by identifier. Packets contained 24–103 words (median 72), within a 220-word budget, and were fixed across paired conditions and replications. Cards excluded reference probabilities, priors, weights, variances, numerical source results, and links identifying reference artifacts. Both evidence arms received node definitions. Input-token budgets were not equalized.

All judgments conditioned on all listed parents and averaged over unspecified circumstances, without fixing omitted circumstances to middle states or assuming independence. Structured prompts framed probabilities as expected counts among 100 hypothetical comparable cases, allowed fractional counts, and requested consistency and normalization checks. These hypothetical cases supplied no observations. A brief evidence note was requested, without collecting private reasoning traces. Influence assumptions were shared between methods, but enforced mathematically only by RNM: nine followed from definitions, 43 had qualitative support, and 40 were uncertain hypotheses. Single-parent means used the direction-aligned parent value without selecting a combination rule.

1.4 Additional outcomes and inference

Full-sample secondary calculations use Jensen–Shannon divergence in bits, mean absolute ordinal-CDF differences over $c - 1$ cutpoints, and smoothed CPT KL divergence. Each uses the same within-node row means and node/family weighting as TV. KL adds 10^{-8} to each state and renormalizes; it is not joint-distribution KL. Older row-weighted KL remains in the historical artifact only. TV remains the primary endpoint. Complete requested row sets and actual request order are available in the replication materials.

Replication blocks, rather than correlated targets, were paired for inference. The paired-t test concerns the mean difference, whereas the exact sign test concerns direction/median under independence. Fresh sessions and shuffled order do not establish normality, independence, or stationarity. Separate Holm families provide exploratory family-wise adjustment, not study-wide confirmatory control. Intervals are pointwise. No smallest effect of interest or equivalence margin was fixed; nonsignificance does not establish equivalence, and significance does not establish practical usefulness.

2 Full-sample complete-table results

All main RQs include 61 nodes in seven BNs and all their CPT rows. Within each replication, the node-weighted score averages rows within nodes and then the 61 nodes equally. The family-weighted sensitivity averages nodes within BNs, BNs within publication families, and the five families equally. Both average the four prompt/evidence combinations equally for the method contrast. These full-sample results supersede the restricted non-anchor endpoint as the paper's primary comparison. The change was made after inspecting earlier results and is exploratory. Original collection, reference probabilities, and predictions are unchanged.

2.1 Method contrasts and replication differences

The two RQ1 weighting contrasts form one Holm family. Intervals are pointwise 95% t intervals over five paired blocks; each sign test has raw $p = 0.0625$ and adjusted $p = 0.125$.

Weighting	Direct	RNM	Δ	95% CI	Raw t p	Holm t p
Equal nodes	0.23171	0.24042	+0.00871	[0.00763, 0.00979]	0.0000239	0.0000478
Equal families	0.19371	0.21888	+0.02517	[0.02114, 0.02920]	0.0000648	0.0000648

Weighting	Block 1	Block 2	Block 3	Block 4	Block 5
Equal families	+0.028551	+0.025946	+0.022504	+0.021120	+0.027729
Equal nodes	+0.008724	+0.009108	+0.007207	+0.009126	+0.009387

2.2 All eight conditions and secondary distances

The following table gives complete-table condition means under each weighting rule; N denotes equal-node weighting and F equal-family weighting. No additional tests are attached to the secondary distances.

Weights	Method	Prompt	RAG	TV	JS bits	Ordinal CDF	KL bits
F	Direct	Base	No	0.19423	0.06605	0.13526	0.24298
F	Direct	Base	Yes	0.19826	0.06870	0.13646	0.25103
F	Direct	Struct	No	0.19102	0.06451	0.13421	0.23813
F	Direct	Struct	Yes	0.19132	0.06571	0.13266	0.24144
F	RNM	Base	No	0.21862	0.08223	0.15265	0.40379
F	RNM	Base	Yes	0.22024	0.08314	0.15172	0.37413
F	RNM	Struct	No	0.21819	0.08265	0.15227	0.42102
F	RNM	Struct	Yes	0.21846	0.08248	0.15137	0.39328
N	Direct	Base	No	0.23267	0.08836	0.16144	0.31892
N	Direct	Base	Yes	0.23489	0.08972	0.16187	0.32147
N	Direct	Struct	No	0.22824	0.08643	0.15834	0.31312
N	Direct	Struct	Yes	0.23105	0.08771	0.15967	0.31609
N	RNM	Base	No	0.23963	0.09305	0.16185	0.37194
N	RNM	Base	Yes	0.24188	0.09369	0.16305	0.36491
N	RNM	Struct	No	0.23891	0.09309	0.16134	0.38091
N	RNM	Struct	Yes	0.24126	0.09348	0.16266	0.37041

2.3 Prompt and evidence contrasts

These eight RQ2 contrasts form a separate Holm family. Prompt is structured minus baseline; RAG is RAG minus no RAG. Each averages the other factor. All exact sign tests fail to reject after adjustment (adjusted values at least 0.5).

Weights	Method	Factor	Δ	95% CI	Raw t p	Holm t p
F	Direct	Prompt	-0.00508	[−0.01126, +0.00111]	0.08502	0.49534
F	Direct	RAG	+0.00216	[−0.00262, +0.00695]	0.27708	1.00000
F	RNM	Prompt	-0.00111	[−0.00677, +0.00456]	0.61643	1.00000
F	RNM	RAG	+0.00094	[−0.00424, +0.00613]	0.64015	1.00000
N	Direct	Prompt	-0.00413	[−0.00595, −0.00232]	0.00319	0.02549
N	Direct	RAG	+0.00252	[−0.00052, +0.00556]	0.08256	0.49534
N	RNM	Prompt	-0.00067	[−0.00490, +0.00356]	0.68195	1.00000
N	RNM	RAG	+0.00230	[−0.00019, +0.00480]	0.06219	0.43535

2.4 Token contrasts across all 61 nodes

The two counters form the RQ3 Holm family. Differences and intervals concern a single replication's method totals; the main paper gives totals over five blocks. Both exact sign tests have raw $p = 0.0625$ and adjusted $p = 0.125$. All 2,444 attempts, including repairs, contribute.

Counter	Mean block Δ	95% CI	Raw t p	Holm t p
Output	-8271.0	[−9416.9, −7125.1]	0.0000366	0.0000732
Input + output	-17914.6	[−39331.4, 3502.2]	0.0809214	0.0809214

Counter	Block 1	Block 2	Block 3	Block 4	Block 5
Output	-6947	-8590	-7873	-8520	-9425
Input + output	-3745	-17486	-4548	-17416	-46378

3 Restricted diagnostics: scope and procedures

The following fitting, non-anchor, and node-omission diagnostics use 25 compression-subset targets (14 Scrum and 11 Chrome) and saved responses from the completed experiment. They do not cover the full sample used for the main RQs. They are post hoc and descriptive. No new LLM requests were made. Original predictions, token totals, statistical tests, and reference tables are preserved. The alternative pipelines below are not substituted into the original eight conditions.

The matched direct-anchor variant uses the saved direct response at the exact configurations queried in the paired RNM request, preserving its row order and LLM-selected family. Pairing fixes node, replication, prompt, and evidence condition. Family selection by anchor fit searches WMEAN, WMIN, WMAX, and MIXMINMAX with the original optimizer and chooses the lowest mean anchor TV without reference access. Stronger search retains the original family and uses 16 initial starts plus the incumbent, 80 sweeps per step, and steps 2^{-2} through 2^{-14}. Seed 42, the 10^{-12} improvement threshold, five-point integration, and influence directions are retained. The incumbent remains a candidate. Expanded bounds change mean log ratios to $[-12, 12]$, minimum/maximum weights to $[1, 100]$, and variance to $[10^{-6}, 4]$, retaining the stronger-search candidate.

Reference-informed diagnostics separately fit complete reference tables. Each node considers all four families with stronger search and frozen signs/bounds, then expanded bounds, then all 2^k parent-direction combinations with expanded bounds. Selection minimizes complete-reference TV; these are in-sample best-found approximations, not predictions on hidden configurations or certified global optima. The no-flip expanded candidate is retained in the

direction search. Family order breaks exact ties. Numerical plateaus can contain different parameters, so low loss does not establish parameter recovery.

4 Alternative pipelines fitted to LLM judgments

The table uses the restricted non-anchor rows and macro-aggregation: equal targets within BNs and equal BN versions within the Chrome family; prompt/evidence cells and five replications receive equal weight. The 14 Scrum and 11 Chrome targets remain separate. Uniform probabilities supply a reference-blind uninformed baseline. No additional p-values are calculated.

Pipeline	Scrum TV	Chrome TV
Original direct	0.3184	0.2223
Original RNM	0.2761	0.2517
Uniform	0.4525	0.3739
Direct anchors, paired RNM family	0.2924	0.2456
RNM anchors, family by anchor fit	0.2787	0.2413
RNM anchors, stronger search	0.2761	0.2517
RNM anchors, expanded bounds	0.2761	0.2530

5 Interpretation of alternative fits

Could the result mainly reflect different LLM answers or an insufficient parameter search? Fitting direct judgments on the paired RNM anchors still improves Scrum relative to the original direct pipeline (0.3184 to 0.2924) and worsens Chrome (0.2223 to 0.2456). Selecting the aggregation family by its fit to LLM anchors narrows Chrome's disadvantage to 0.2413, but does not reverse it. Stronger searches and wider parameter bounds make little difference. These post-hoc checks support a mismatch in the tested reconstruction pipeline rather than a simple shortage of optimization effort; they do not isolate every component.

Influence assumptions also matter. Four Chrome targets exhibit mean-direction reversals and six exhibit stochastic-order reversals under the specified parent alignment. Scrum has two targets with reversals. Even fitting directly to complete reference tables yields a higher best-found error for Chrome than Scrum (0.1592 versus 0.0560); allowing wider bounds and reversed directions lowers these to 0.1322 and 0.0260. This reference-informed diagnostic supports a compatibility explanation, without establishing a global optimum or a lower bound on RNM error.

6 Reference-informed approximation diagnostics

The following table reports complete-table TV, including fitted configurations. These values are not comparable to held-out predictive validation or to a mathematical error floor.

Reference-informed search	Scrum TV	Chrome TV
All families, stronger search	0.0560	0.1592
Expanded bounds	0.0559	0.1589
Expanded bounds and direction enumeration	0.0260	0.1322

7 Non-anchor diagnostic metrics and replication differences

Metric	Scrum D	Scrum RNM	Chrome D	Chrome RNM
TV	0.3184	0.2761	0.2223	0.2517
Jensen–Shannon (bits)	0.1331	0.1066	0.0905	0.1104
Ordinal CDF	0.1887	0.1588	0.1809	0.2000

The five original replication differences (RNM minus direct TV) are:

Replication	Scrum	Chrome
1	-0.04538	+0.03410
2	-0.03414	+0.02773
3	-0.04732	+0.02682
4	-0.04155	+0.02935
5	-0.04314	+0.02907

8 Illustrative conditional distributions

For each stratum, we select the target nearest the unweighted median per-target RNM-minus-direct non-anchor TV effect, breaking ties by identifier. We show its all-middle aligned anchor and average the 20 saved distributions per method. This transparent rule does not select the largest benefit or harm. Mean probability vectors are illustrative; their TV need not equal mean trial TV.

8.1 All contributors

Number of distinct component contributors.

Parents: Component changes = [11, 51); Distinct reviewers = [3, 7).

Child state order: [1, 3), [3, 5), [5,102]. The target's mean non-anchor effect is +0.0193. All 20 RNM trials select WMEAN.

Distribution	State 1	State 2	State 3
Reference	0.356	0.421	0.223
Direct	0.261	0.358	0.382
Raw RNM	0.213	0.396	0.391
Fitted RNM	0.231	0.537	0.231

At this row, mean trial TV is direct: 0.1587, raw rnm: 0.1691, fitted rnm: 0.1318. These anchor illustrations do not establish a universal pattern for non-anchor rows.

8.2 Team Orientation

Mutual trust, respect and motivation to prioritize team goals over isolated individual goals.

Parents: Personal Attributes = Medium; Expertise = Medium.

Child state order: Low, Medium, High. The target's mean non-anchor effect is −0.0306. All 20 RNM trials select WMEAN.

Distribution	State 1	State 2	State 3
Reference	0.053	0.927	0.020
Direct	0.237	0.501	0.263
Raw RNM	0.206	0.585	0.210
Fitted RNM	0.207	0.586	0.207

At this row, mean trial TV is direct: 0.4260, raw rnm: 0.3425, fitted rnm: 0.3414. These anchor illustrations do not establish a universal pattern for non-anchor rows.

9 Qualitative evidence packets

All 61 retained nodes have nonempty packets containing 1–3 cards. Card text totals 24–103 whitespace-separated words per node, with median 72. The same node packet is reused across methods, prompt variants, and replications. The following are the complete cards for the two illustrative targets. Both evidence arms also receive the child and parent definitions.

9.1 All contributors

Component history. Changes are aggregated at component-directory level. Size, change volume and contributor counts describe different aspects of development activity; one does not uniquely determine another.

Contributor roles. Contributors are classified by their share of component changes. Minor and major contributors partition the contributor set. Holding total contributors fixed, a larger minor-contributor count leaves fewer major contributors.

Review measures. Review activity can include discussion, approval and revisions of a patch. A patch revision during review is not itself a confirmed post-release defect. Author-only approval and review without discussion describe distinct practices.

9.2 Team Orientation

Mutual support. Team orientation involves mutual respect and giving team goals priority. Constructive relationships and confidence in colleagues help members work together toward those goals.

Knowledge coverage. Expertise includes collective knowledge and overlapping capabilities. Being able to perform parts of another member's work supports flexibility when circumstances demand coverage.

Assessment scope. The assessment is intended to support retrospective discussion. Project-specific practices and metrics can be added to a general model. A contextual ordinal rating does not have a universal numerical threshold across organizations.

10 Target-level non-anchor diagnostic differences

10.1 Scrum

Network	Target	Direct	RNM	Δ
Scrum	Teamwork Quality	0.3050	0.3095	+0.0044
Scrum	Increment quality	0.3306	0.3510	+0.0204
Scrum	Customer satisfaction	0.1131	0.0694	-0.0437
Scrum	Cohesion	0.2471	0.1018	-0.1452
Scrum	Self-Organizing	0.3491	0.3487	-0.0004
Scrum	Collaboration	0.3671	0.2777	-0.0894
Scrum	Team Orientation	0.3869	0.3563	-0.0306
Scrum	Communication	0.4366	0.3707	-0.0660
Scrum	Coordination	0.1500	0.1284	-0.0216
Scrum	Continuous backlog refinement	0.2454	0.1708	-0.0746
Scrum	Sprint execution quality	0.3522	0.3316	-0.0206
Scrum	Requirements volatility	0.5307	0.4614	-0.0692
Scrum	Product backlog quality	0.3366	0.3062	-0.0304
Scrum	Sprint planning quality	0.3074	0.2821	-0.0253

10.2 Chrome

Network	Target	Direct	RNM	Δ
chrome_40	Distinct reviewers	0.1189	0.1419	+0.0229
chrome_40	Major contributors	0.0676	0.0728	+0.0052
chrome_40	Minor contributors	0.3389	0.3860	+0.0471
chrome_40	Prior defects	0.1728	0.1240	-0.0487
chrome_40	Patch revisions during review	0.3301	0.3146	-0.0156
chrome_40	Self-approved changes	0.2133	0.2219	+0.0086
chrome_42	All contributors	0.2139	0.2332	+0.0193
chrome_42	Major-author ownership	0.4344	0.4255	-0.0089
chrome_42	Major contributors	0.0000	0.1565	+0.1565
chrome_42	Minor contributors	0.3696	0.4561	+0.0866
chrome_42	Self-approved changes	0.1706	0.1949	+0.0243

These target means describe this sample; targets and rows are not independent experimental replications.

11 Version differences and leave-one-node-out diagnostics

These post-hoc descriptive checks reuse the restricted non-anchor scores, without refitting or new hypothesis tests. Each omission removes one compression-subset target and re-averages the remaining targets within its BN; the two Chrome versions retain equal weight. Prompt/evidence cells and the five replications retain equal weight. Predictions, references, non-anchor row sets, and the restricted diagnostic sample remain unchanged.

The version-level means are:

Network	Targets	Direct TV	RNM TV	Δ
Chrome 40	6	0.2069	0.2102	+0.0033
Chrome 42	5	0.2377	0.2933	+0.0556
Scrum	14	0.3184	0.2761	-0.0423

Chrome's aggregate disadvantage is concentrated in Chrome 42. No individual omission changes the original direction in any of the five replication blocks. The ranges below are descriptive minima and maxima across omissions, not confidence intervals.

Stratum	Omissions	Minimum Δ	Maximum Δ
Scrum	14	-0.0471	-0.0344
Chrome	11	+0.0168	+0.0375

Major Contributors in Chrome 42 has per-target difference +0.1565 and original macro-weight 0.1. Its weighted contribution is +0.01565, or 53.2% of the net Chrome contrast; this is an arithmetic contribution, not explained variance. Omitting it and reweighting within Chrome 42 leaves Δ = +0.0168, positive in every block. The node remains in all original analyses.

11.1 Scrum single-node omissions

Omitted target	Network	Remaining macro Δ
Teamwork Quality	Scrum	-0.0459
Increment quality	Scrum	-0.0471
Customer satisfaction	Scrum	-0.0422
Cohesion	Scrum	-0.0344
Self-Organizing	Scrum	-0.0455
Collaboration	Scrum	-0.0387
Team Orientation	Scrum	-0.0432
Communication	Scrum	-0.0405
Coordination	Scrum	-0.0439
Continuous backlog refinement	Scrum	-0.0398
Sprint execution quality	Scrum	-0.0440
Requirements volatility	Scrum	-0.0402
Product backlog quality	Scrum	-0.0432
Sprint planning quality	Scrum	-0.0436

11.2 Chrome single-node omissions

Omitted target	Network	Remaining macro Δ
Distinct reviewers	Chrome 40	+0.0274
Major contributors	Chrome 40	+0.0292
Minor contributors	Chrome 40	+0.0250
Prior defects	Chrome 40	+0.0346
Patch revisions during review	Chrome 40	+0.0313
Self-approved changes	Chrome 40	+0.0289
All contributors	Chrome 42	+0.0339
Major-author ownership	Chrome 42	+0.0375
Major contributors	Chrome 42	+0.0168
Minor contributors	Chrome 42	+0.0255
Self-approved changes	Chrome 42	+0.0333

12 Complete-table family means and anchor fitting contrasts

Complete-table TV averages are Scrum 0.2605 direct versus 0.2334 RNM; Chrome 0.2029 versus 0.2278; APS 0.0964 versus 0.1155; SESSI 0.1160 versus 0.1143; and software logs 0.2927 versus 0.4034. The equal-family external SE average is 0.1770 versus 0.2153. Excluding SESSI and reweighting gives 0.1973 versus 0.2489. The family means collectively cover all 61 targets, while the external-only aggregate is a restricted diagnostic. The primary full-sample endpoint includes Scrum and all external BNs together.

On identical anchor rows of compression-subset targets, fitting changes Scrum TV by −0.0028 (pointwise 95% CI $[-0.0053, -0.0003]$, raw and Holm t $p = 0.0352$), and Chrome TV by +0.0503 (CI $[0.0430, 0.0576]$, raw $p = 4.46 \times 10^{-5}$, Holm $p = 8.93 \times 10^{-5}$). Both adjusted exact sign-test values are 0.125. These are historical exploratory tests with their own two-contrast Holm family, outside the revised RQ families. They use five replication blocks and do not decompose non-anchor error additively.

Table 1. Diagnostics on identical anchor configurations of the 25 compression-subset targets. Direct, raw RNM, and fitted RNM errors are against the reference. Fit residual compares fitted and raw RNM judgments. All use the restricted diagnostic weights.

Quantity	Scrum	Chrome
Direct anchor TV	0.2914	0.2507
Raw RNM anchor TV	0.2790	0.2491
Fitted anchor TV	0.2762	0.2994
Fit residual TV	0.0678	0.1253
Reconstructed non-anchor TV	0.2761	0.2517